Integrated CAD Approach for Generating Assembly Sequence Plans Based on Collision Study Results

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Abstract. Given its economical and technical benefits, the assembly sequence planning remains an indispensable process during the life-cycle of a mechanical product. Many approaches and methods had been proposed to design and optimize the assembly sequences of a mechanical product. Moreover and in order to be more efficient, many researchers adopted integrated solutions to the CAD systems. In this paper an approach using a collision study and permitting the generation of the assembly sequences planning of a mechanism is proposed. The collision test is performed by the use of both the API and the interference test tool of the CAD software. Driven by the obtained results and the collected data from the CAD assembly file, the developed algorithm determines only feasible sequences.

Keywords: ASP, assembly model, integrated design, collision study.

1 Introduction

The assembly sequence planning (ASP) is an innovative concept which allows the improvement of the assembly design. The ASP offers many economical and technical advantages such as the control of the production costs and the improvement of the maintenance processes. Given its advantages, the ASP problem has taken its place into the life-cycle of mechanical products and it has been the subject of many research works.

In the literature, several works have been proposed to generate and/or to optimize the assembly plans using various methods such as graphs, questions, heuristics and geometric reasoning based approaches. The methods based on geometric reasoning, employ generally an interference analysis to carry out or to check the feasibility of the assembly plans. Some relevant examples of contributions about this object are as follows: Dini and Santochi are the firsts to propose an approach which uses interference matrices (Dini and Santochi 1992). This method, allows the detection of subassemblies based on the product's interference, contact, and connection matrices. Dini and Santochi modelled the interferences through six

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matrices according to ±X, ±Y and ±Z axes. Huang et al. proposed an improved method to generate disassembly sequences based on the interference matrices (Huang et al. 2002). Ciszak proposed an approach which combines the theory of graphs, heuristics algorithms and collision matrix to generate and simulate ASP (Ciszak 2012). Liu et al. developed an integrated approach to CAD system which carries out the disassembly sequences using the interference matrix and calculation of the disassembly constraint degree of components (Liu et al. 2013). The proposed method allows the inspection of the manoeuvrability of the assembly tool, however, it's limited to X,Y and Z axes. Viganò and Gómez proposed an automatic method based on the contacts and interferences data acquired from the assembly model (Viganò and Gómez 2013). The proposed approach generates at least a feasible solution. The cited contributions above are based on Dini and Santochi interference matrices which are restricted to the three axes of the coordinate system. Hu et al. proposed an integrated method which determines the disassembly paths by using the information extracted from the mating features of components (Hu et al. 2002). This method uses an interference test to check the non-presence of collision during the disassembly of components.

In this work, an automatic approach integrated to CAD system which generates feasible assembly sequences based on collision study is presented. The different stages of the developed approach are explained through an illustrative example.

2 Collision Study

The collision study is performed in order to describe spatial relations between components by identifying any obstacles which may be present in the assembly path. In this work, each component of the mechanical assembly is moved by a translational displacement to simulate its disassembly. During this movement, the analyzed component may collide with other elements of the mechanism. In this case, the names of the interfering elements are retrieved and saved. When the component arrives at the final point of displacement, it returned to its initial position in order to be analyzed in another direction. This procedure is repeated with all components of mechanism.

Figure 1 shows an illustrative example used to explain the developed algorithm. The considered mechanism is an air starting valve composed of ten components.

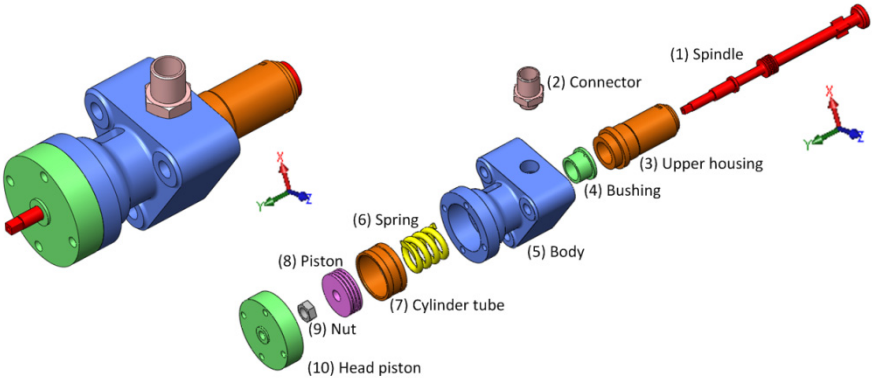

Fig. 1 Illustrative example: Air starting valve

2.1 Assembly Directions

The collision study is performed according to defined directions which are specifics to each component. In the literature, approaches based on interference study use interference matrices in which the components are tested along the ±X, ±Y, and ±Z axis of the coordinate system of the mechanical product. However, this method may constitute a limitation to generate ASP especially with the complex geometry mechanism. As an example, the component "Connector" of the illustrative mechanism, can be assembled along the Δ axis which is an inclined axis specified by the director vector $\vec{U}$ (Fig. 2).

Where

$\vec{U} = 0{,}92\vec{x} + 0{,}39\vec{z}$

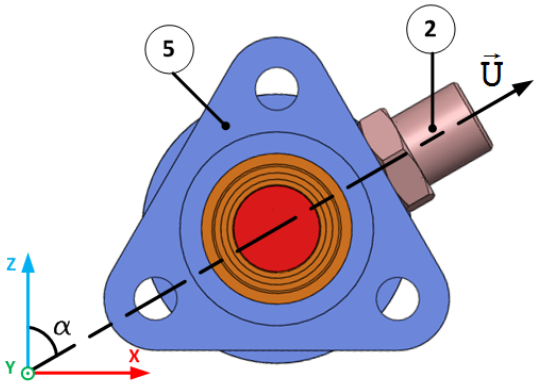

Fig. 2 Example of inclined assembly direction

To ensure accurate results and to be compatible with any component having specific shapes, the collision test is carried out along directions deduced from the assembly constraints "mates" of the CAD model. The algorithm of identification of assembly directions is illustrated by the flowchart in the figure 3. The main idea of the proposed algorithm is the analysis of the contact type and the director vector of assembly constraints.

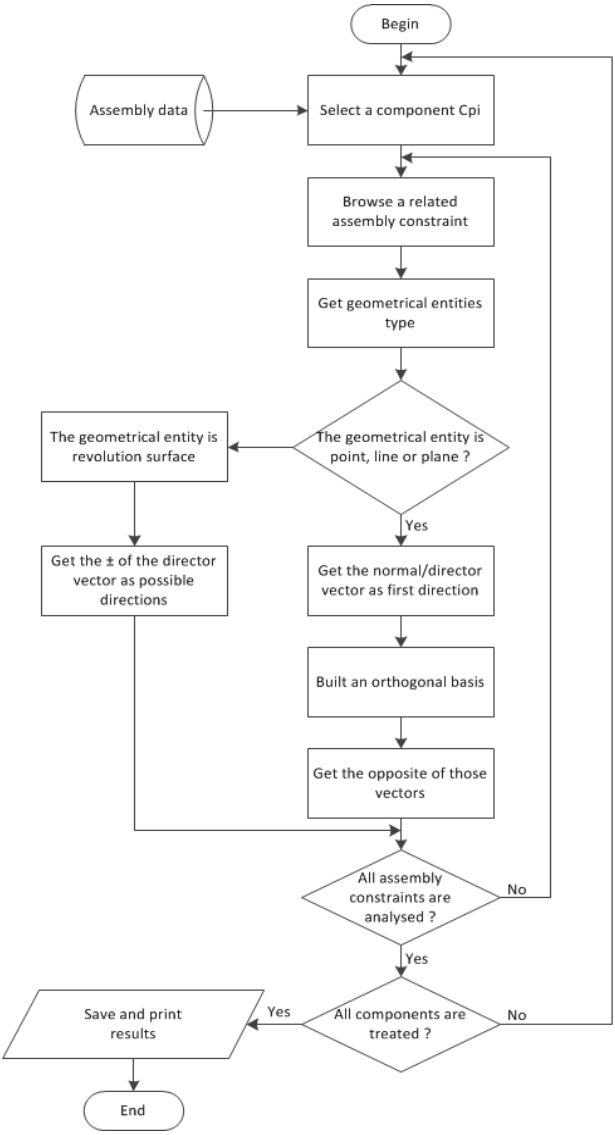

Fig. 3 Flowchart of directions identification algorithm

2.2 Collision Results

Given the diversity, each component can have a different set of possible assembly directions according to its geometric position. The results of the approaches based on Dini and Santochi method, are presented in six interference matrices. Differently to the previous cited approaches, the developed approach presents the results as tables. Each table, representing the collisions results for a specific component, contains the interfering elements according to directions of the analysed component. Table 1 presents an example of the connector collision study of illustrative example. As it can be noticed, the connector collides with body, spindle, bushing and upper housing according to Δ axis.

Table 1 Collision results of component "Connector"

Component name	Connector	
Direction [i, j, k]	[0,92, 0, 0,39]	[-0,92, 0, -0,39]
Interfering components names		Body Spindle Bushing Upper housing

3 Assembly Sequence Generation Process

To generate ASP, the developed algorithm begins by the calculation of a score to each component. These scores will be used for identification of the base element and to rank components. Subsequently, the algorithm searches the sequences based on geometric reasoning which uses the contact relationship between components and the collision results.

3.1 Scores Components Calculation

Given the lack of information about the precedence relationships, a score calculation for each component was proposed. The component score is calculated via the volume, area and the number of relationships.

The calculated scores serve in the first step to define the base component on which most elements will be assembled. In fact, the base component must have the higher score and can't be a fastener element. In a second stage, those scores will be used by the ASP algorithm for the components' ordering. The algorithm of scores calculation and base component identification is presented by the flowchart in figure 4.

For a component Cp_i, the score Sc_i is calculated according to formula (1)

$$Sc_i = 0{,}35.\frac{V_i}{V_a} + 0{,}35.\frac{A_i}{A_a} + 0{,}3.\mathrm{Nb} \quad (1)$$

where

- V_i and A_i are respectively the volume and area of the component Cp_i,
- V_a and A_a are respectively the average volume and area of components constituting the assembly,
- Nb represents the number of relationships of component Cp_i

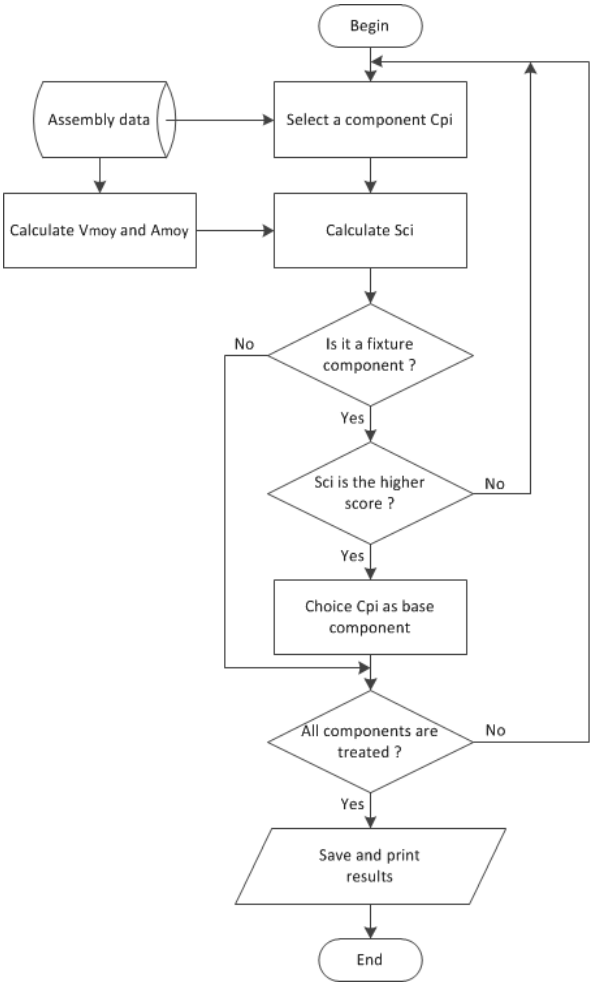

Fig. 4 Algorithm of scores components **calculation and base identification**

3.2 Assembly Sequences

Once the collision study is accomplished the ASP algorithm, presented by the flowchart in figure 5, begins by the scores calculation process. Thereby, all components are sorted in decreasing order according to their scores then the base component is identified. Therefore, the algorithm selects the related components

with the base component respecting the score order. For each related component, the algorithm analyses the possible assembly directions (presented in section 2.1). If the checked direction doesn't present an assembled element, then, it's considered as an assembly direction for the present component. Once all component related to the base are assembled, the algorithm denotes an assembled component as a new base component by respecting the descending order of scores. This procedure is repeated until all components are processed.

In some cases, a component with lower score must be assembled before another having a higher one. When the algorithm detects the impossibility of assembling a component, it changes the order of the concerned element and re-executes the procedure until solving the problem.

On the illustrative example, the calculated scores are presented in table 2. The base component is the body (5). The algorithm begins by assembling components related to this element by respecting the descending order of scores. During this process, a case of impossible assembly was detected. A demonstration to solve this problem is explained as follows:

Step1: (5)

Step2: (5) → (3)

Step3: (5) → (3) → (10)

Step4: (5) → (3) → (10) → (7): Component (7) can't be assembled ⇒ Impossible sequence ⇒ Permutation of (7) with (10) and re-execute the ASP algorithm.

Step5: (5) → (3) → (7) → (10) → (6): Component (6) can't be assembled ⇒ Impossible sequence ⇒ Permutation of (6) with (10) and re-execute the ASP algorithm.

Step6: (5) → (3) → (7) → (6) → (10) → (4) Component (4) can't be assembled ⇒ Impossible sequence ⇒ Permutation of (4) with (10) and re-execute the ASP algorithm.

Step7: (5) → (3) → (7) → (6) → (4): Component (4) can't be assembled ⇒ Impossible sequence ⇒ Permutation of (4) with (6) and re-execute the ASP algorithm.

Step8: (5) → (3) → (7) → (4): Component (4) can't be assembled ⇒ Impossible sequence ⇒ Permutation of (4) with (7) and re-execute the ASP algorithm.

Step9: (5) → (3) → (4): Component (4) can't be assembled ⇒ Impossible sequence ⇒ Permutation of (4) with (3) and re-execute the ASP algorithm.

Step10: (5) → (4) → (3) → (7)

After a few iterations, the final result of the air starting valve is presented by figure 6.

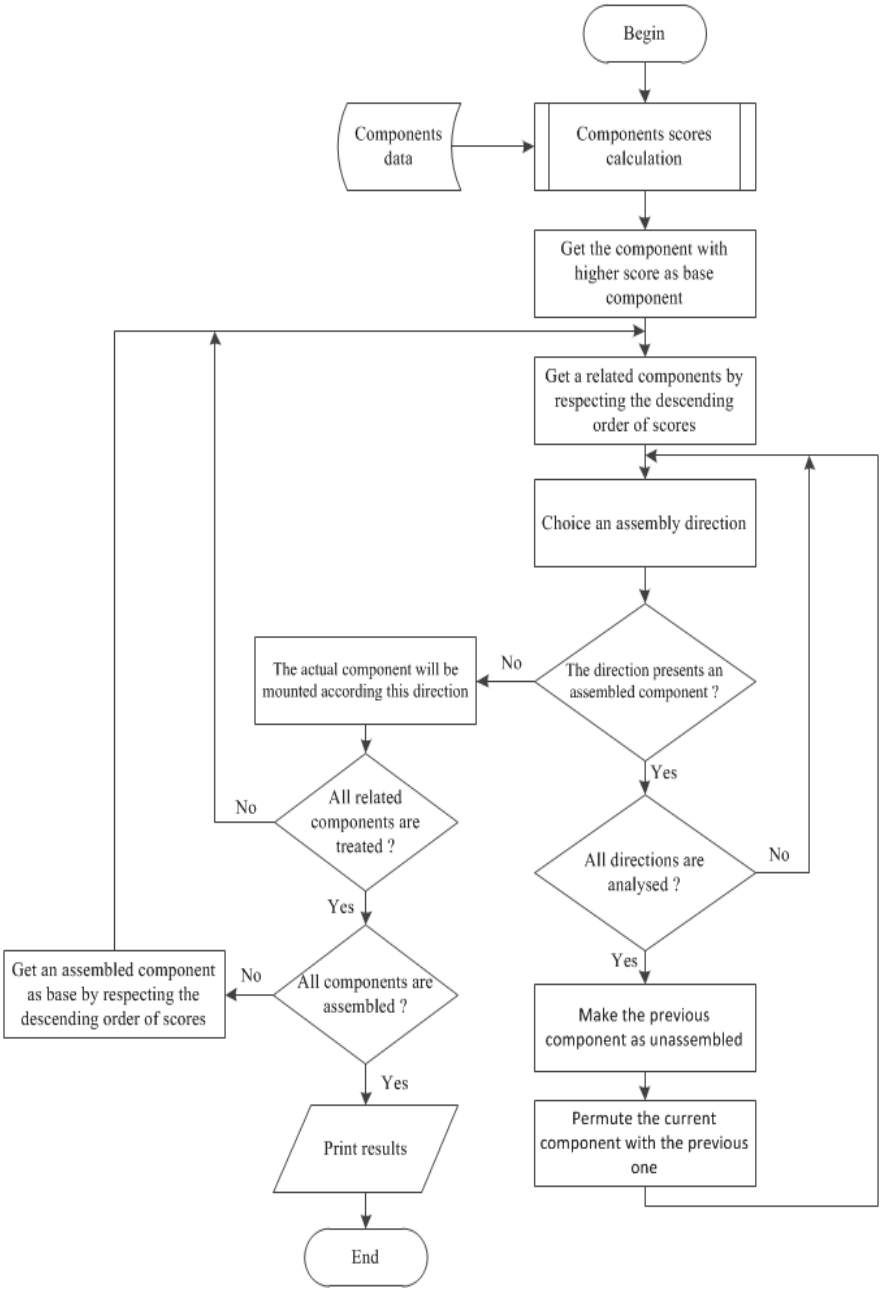

Fig. 5 General algorithm of assembly sequence generation

Table 2 Calculated scores of the illustrative example

	Component	Score Sc
1	(5) Body	4,981901535
2	(1) Spindle	1,941839493
3	(8) Piston	1,518659854
4	(3) Upper housing	1,403503685
5	(10) Head piston	1,298692677
6	(7) Cylinder tube	0,864483507
7	(6) Spring	0,861620095
8	(4) Bushing	0,735552781
9	(9) Nut	0,638321465
10	(2) Connector	0,555424907

	Component	Direction	Score
	(5) Body		4,98190153
1	(4) Bushing	[0, 1, 0]	0,73555278
2	(3) Upper housing	[0, 1, 0]	1,40350369
3	(7) Cylinder tube	[0, -1, 0]	0,86448351
4	(6) Spring	[0, -1, 0]	0,86162009
5	(2) Connector	[-0,92, 0, -0,39]	0,55542491
6	(1) Spindle	[0, 1, 0]	1,94183949
7	(8) Piston	[0, -1, 0]	1,51865985
8	(9) Nut	[0, -1, 0]	0,63832147
9	(10) Head piston	[0, -1, 0]	1,29869268

Fig. 6 ASP results of the air starting valve example

4 Conclusion

In this work, an automatic approach for generating assembly sequences based on collision study and relationships between components was presented. The collision study permits the detection of any eventual interference during the assembly of components. Then, an algorithm of ranking components and determining the assembly directions was performed. Finally, the presented approach does not need prior precedence knowledge and it offers the advantage of generating only possible sequences.

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